Higher National Diploma in Engineering (Mechanical/Electrical)
Forth Year, First semester Examination - 2023
ME 3227 – Industrial Engineering Management

Instructions for Candidates:
Answer any 5 questions.

No. of questions : 06
No. of pages : 04
Time allowed : 03 hours

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1)

a) What are the advantages of using a model in comparison to experimenting with the real situation? (04 marks)

b) A meat factory produces two grades (A and B) of hot dog mixtures in 5kg packs required chicken, beef and flour as three ingredients. Resource requirements and profits are tabulated below.

Type of hot dog grade 5kg packs	Resource requirement per each grade of hot dog (for a 5kg pack)			Profit from each hot dog 5kg pack
	Chicken	Beef	Flour	
A	1kg	3kg	1kg	Rs. 500/=
B	1kg	1kg	3kg	Rs. 600/=

Maximum availability of chicken per day is 200kg, beef per day is 300kg. According to supplier agreement flour should be bought at least 300kg or more than that per day.

Assuming that the food maker wants to maximize profit,

i) Formulate Linear Programming model. (06 marks)

ii) By using the graphical method, find the number of each grade of hot dog 5kg packs to be produced daily for maximizing the profit. (08 marks)

iii) What is the maximum profit that the company can achieve within a day? (02 marks)

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2)

A) a) Define following terms related to economics. (06 marks)

i) Law of marginal utility ii) Demand iii) Supply

b) Discuss the disadvantages of oligopoly market structure giving suitable examples from Sri Lankan market. (04 marks)

B) a) Define following terms. (04 marks)

i) Variable cost ii) Fixed cost

b) Selling price for 1kg of cake is = Rs.1,200.00 and variable expenses per 1kg of cake are as follows:

Flour = Rs.65.00 Eggs = Rs.120.00 Butter = Rs.150.00
Sugar = Rs.65.00 Labour = Rs.100.00
Variable overheads = Rs.100.00

Related Fixed Costs: Bakery rent = Rs.12,000.00 per month

Find the number of kgs of cakes to be produced to break even the business.

(06 marks)

3)

a) A company is going to undertake a project that require an initial investment of Rs.2,400,000.00 and it generates annual net income of Rs.800,000.00 for next 6 consecutive years. What is the simple payback period of the project? (04 marks)

b) A Businessman has a building to lease or rent and received proposals from 2 companies. He has to spend Rs.50,000.00 for renovation of the building before rent and then Rs.20,000.00 for the annual maintenance at end of each year for next 5 years.

Option 01 – Kings company proposed to pay him Rs.200,000.00 initially and Rs.100,000.00 at the start of 2nd year. For next 3 years, the amount they pay will be increased by 10% every year compared to the previous year.

Option 02 – Queens company proposed to pay him Rs.100,000.00 initially and Rs.125,000.00 at the start of each year for next consecutive 4 years.

3) b)

i) Calculate the yearly Net cash flows of the two alternatives separately. (06 mark

ii) Calculate the NPV for both options at a rate of 10% interest. (08 mark

iii) Which alternative will be the best option for the businessman and why do yo recommend it? (02 mark

4)

a) What are the main 5 phases of a project life cycle? (05 marks)

b) Discuss the importance of Gantt chart as a project management tool. (05 marks)

c) What are the main 4 types of organizational structures? (04 marks)

d) Selecting any one type of those organizational structures, discuss the applicability of it for an organization, referring to suitable examples. (06 marks)

5)

a) State any 5 dimensions of quality. (05 marks)

b) Distinguish the difference between quality assurance and quality control. (05 marks)

c) Write short notes on following topics. (10 marks)

i) Use of different leadership styles by a manager
ii) Importance of human resource planning for an organization

6)

a) Discuss the applicability and advantages of fixed position layout for special manufacturing processes, with reference to examples.

(04 marks)

b) The SL motor manufacturing company produces 12 motors per day and the manufacturing facility operates for 10 hours per day. Manufacturing of motors consists of following activities with the given time allocation.

Task	A	B	C	D	E	F	G	H	I	J	K
Predecessor	-	-	A	A	B	D,E	C	G,F	H	H	I,J
Task time (min)	30	18	15	15	20	25	20	10	35	22	26

i) Draw the network diagram. (06 marks)

ii) Compute the cycle time per workstation in minutes. (02 marks)

iii) Compute the minimum number of workstations. (02 marks)

iv) How would you allocate the tasks into workstations? (04 marks)

v) What is the efficiency of line balancing? (02 marks)